

Javad Komijani

Computational Scientist & ML Researcher

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Professional Summary

Computational scientist and machine learning researcher with 10+ years of experience in scientific computing and 5 years in generative modeling. Specialized in PyTorch-based diffusion models, normalizing flows, and differentiable programming. Strong background in applied mathematics, statistical modeling, and scientific software engineering.

Technical Skills

Programming	Python, C++, Cython, Bash
ML/AI	PyTorch, Diffusion Models, Normalizing Flows, Generative Modeling, Distributed Training
Mathematics	Differential Equations, Linear Algebra, Optimization, Statistical Modeling
Tools	Linux, Git, SQL, Pandas
Scientific	NumPy, SciPy, Scikit-learn, HPC, Parallel Computing, Monte Carlo Methods

Professional Experience

- 2021–Present **Senior Postdoctoral Researcher, ETH Zurich**
- Developed a novel score-matching framework for diffusion models and normalizing flows on Lie groups, and implemented installable GPU-based distributed training infrastructure for large-scale lattice simulations.
 - Developed installable PyTorch extension packages for differentiable linear algebra and ODE solvers.
 - Developed and contributed ML modules to the interTwin (EU interdisciplinary Digital Twin Engine).
 - Supervised PhD and Master's students and taught quantum mechanics.
- 2015–2021 **Postdoctoral Researcher, TU Munich / Glasgow / Tehran**
- Developed models and applied Bayesian inference & Monte Carlo simulations to large-scale datasets.
 - Developed scalable lattice-QCD simulation pipelines in HPC environments.
 - Collaborated internationally in computational and mathematical physics projects.

Open-Source Scientific ML Packages

normflow	Installable PyTorch package for normalizing flows on Lie groups & lattice gauge theory.
diffusion	Installable PyTorch package for diffusion-based generative modeling on Lie groups & gauge theories.
torch_linalg	Installable PyTorch package for differentiable SVD and eigendecomposition of unitary matrices.
torch_solve	Installable PyTorch package implementing adjoint-based differentiable ODE solvers for scalar and Lie-group-valued systems.

Education

- 2015 **PhD in Physics, Washington University in St. Louis**
2009 **MSc in Electrical Engineering, University of Tehran**

Publications

A complete list of over 25 peer-reviewed publications can be found at [this link](#).

Languages

English	Fluent
German	Elementary